

M004 · Sensing the World — English Worksheet

Topic: Sensing the World · **Course:** Functions & Games · **Time:** about 45 minutes

This lesson your agent **senses** the world before it acts. It can detect a block **ahead**, check what is **below**, and act differently for each block type.

1 · Predict

```
if agent detects block ahead:
    turn left
otherwise:
    move forward
```

There is a stone wall right in front of the agent. What does it do? What if the path ahead is open?


```
if block below is lava:
    place block below
    move forward
otherwise:
    move forward
```

The agent is standing at the edge of a lava pool. What does it do before moving? Does it fall in?


```
repeat 5 times:
  if block below is dirt:
    place flower
  if block below is sand:
    place cactus
  move forward
```

The agent walks over: dirt, sand, dirt, stone, sand. What gets planted on each step?

2 · Spot the Bug 🐛

Each block below is broken. Rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — The agent should place a safe block when the block **below** is lava. But it never notices the lava under its feet and falls in.

```
if agent detects block ahead:
    place block below
move forward
```

Hint: the lava is **below**, but the check looks **ahead**.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent should plant a flower **only** when it is standing on dirt. But it plants a flower on **every** step, even on stone.

```
repeat 5 times:
    if block below is air OR block below is dirt:
        place flower
    move forward
```

Hint: above the ground, the block ahead of the agent is almost always air — so this check is almost always true.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent should cross the field safely: cover lava when it sees it, and walk when the ground is safe. But this code **only** handles the lava case — on safe ground the agent does nothing and gets stuck.

```
repeat 8 times:  
  if block below is lava:  
    place block below  
    move forward
```

Hint: add an `otherwise` so the agent still moves on safe ground.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Send your agent across the danger field: check the block **below** and the block **ahead**, and act differently for **two** block types.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.